

Manasa A

[+91 7338033513](tel:+917338033513) | manasa.a1205@gmail.com | linkedin.com/in/manasa-a-868722304 | github.com/manasa2212

Summary

Undergraduate Computer Science student with strong interest in software development, backend systems, and web technologies. Proficient in Java, Python, and database development with hands-on experience building IoT and software projects. Familiar with REST APIs, SQL databases, Git, and Linux environments. Passionate about developing scalable applications and continuously learning new technologies.

Education Qualification

- Pursuing an undergraduate in the stream of Computer Science and Engineering at BNMIT College with a 9.59 CGPA and graduating in 2027
- Completed PU education with a focus on science, achieving an 92.76% at Jnana Jyothi PU College in 2022
- Completed SSLC education, achieving 93.76% at Jnana Jyothi High School in 2020

Projects

- **Smart Irrigation System**
Developed an IoT-based smart irrigation system using Raspberry Pi to monitor soil moisture and automatically control water pumps. Integrated real-time weather API data to optimize irrigation and prevent overwatering. Implemented sensor data processing and automation logic using Python.
- **Password Strength Audit**
Developed a password auditing tool in **Linux** to analyze password strength using **dictionary and brute-force techniques**. Used security tools to identify weak credentials and demonstrate common password vulnerabilities. Applied cybersecurity best practices for secure authentication systems.
- **ML based to refine-captcha**
Designed a machine learning model to detect human vs bot interaction based on mouse movement patterns and user behavior. Built a backend classifier using Python and trained models on behavioral datasets to improve web security. The solution provides a privacy-friendly alternative to traditional CAPTCHA systems.

Skills

Technical Skills

- **Programming Languages:** C(Basic), Java, Python
- **Database:** MySQL, MongoDB
- **Web Development:** HTML/CSS
- **Tools and Platforms:** VS Code, GitHub, Linux, RaspberryPi OS

Soft Skills

- Teamwork
- Communication
- Adaptability
- Creativity

Internships

- Completed an **Penetration Testing: Concepts, Tools and Techniques** internship at Ganaka Praudayogikee (2/06/2025 – 27/06/2025), learned penetration testing methodologies and applied industry-standard tools to identify and analyze system vulnerabilities.
- **Virtual Internship – Web Development | SkillCraft Technology (Aug 1 – Sep 1, 2025)**
Completed a one-month virtual internship in web development at Skill Craft Technology. I learned to build and design websites using HTML, CSS, and JavaScript, and gained practical experience in creating responsive and easy-to-use web pages.

